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FACE MASK DETECTION USING DEEP LEARNING

A Mini Project Report submitted in partial fulfillment of the requirements for the award of the degree of

MASTER OF SCIENCE IN COMPUTER SCIENCE

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CERTIFICATE

The research work embodied in the present Dissertation entitled "**Face Mask Detection Using Deep Learning**" is a bonafide work done by **A.SEKAR (Register Number: 24081236103911113)**, Department of Computer Science, Sri Paramakalyani College, Alwarkurichi, Tenkasi - 627412 has been carried out in partial fulfillment of the award of the Degree of **Master of Science in Computer Science** during the academic year 2025 – 2026.

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DECLARATION

I hereby declare that this Dissertation entitled "**Face Mask Detection Using Deep Learning**" submitted by for the award of the Degree of **MASTER OF SCIENCE IN COMPUTER SCIENCE,** is the result of my original and independent project work carried out under the guidance of **F.Vasantha Dyana M.sc., M.Phil.,** Assistant Professor, Department of Computer Science, Sri Paramakalyani College, Alwarkurichi.

I also declare that this written submission presents my idea in my own words, and where others' ideas or words have been included, I have adequately cited the referenced the original sources.

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ABSTRACT

(COVID-19) pandemic and its several variants have developed new habits in our daily lives. For instance, people have begun covering their faces in public areas and tight quarters to restrict the spread of the disease. However, the usage of face masks has hampered the ability of facial recognition systems to determine people's identities for registration authentication and dependability purpose. This study proposes a new deep-learning-based system for detecting and recognizing masked faces and determining the identity and whether the face is properly masked or not using several face image datasets. and the most effective protective strategy is wearing a face mask in public places (WHO). According to reports, wearing a facemask in general settings minimizes the chance of transmission. This paper describes and construct an ensemble of Deep Learning and Machine Learning algorithms to recognize persons without face masks and send an email alert when the count threshold crosses. The facemask dataset consists of images of people with and without masks. Our objective is to take advantage of existing cameras and detect persons wearing masks and avoid virus transmission. scaling to increase the size of the training dataset and improve the performance of the model. Our face mask detection system is designed to work in real world scenarios where lighting conditions and occlusions can be challenging. The MobileNetV2 architecture is also used, it's computationally efficient and thus making it easier to deploy the model to embedded systems. One of the most effective measures is the use of face masks. According to reports, wearing a facemask in general settings minimizes the chance of transmission. To solve this challenge, we needed a reliable mask monitoring system. In this project, we propose a face mask detection system that utilizes deep learning classification algorithms to detect persons whether an individual is wearing a face mask or not. The facemask dataset consists of images of people with and without masks and the

dataset is taken from Kaggle. This paper aims to use face mask detection system is designed to work in real world scenarios to improve public health safety measures during ongoing and future health crises.

CHAPTER-1

1.1 INTRODUCTION

The COVID-19 pandemic has highlighted the importance of personal safety measures, especially the use of face masks to prevent the spread of infectious diseases. As wearing face masks became mandatory in public places, the need for automated systems to monitor mask compliance grew rapidly. Manual monitoring is not only labor-intensive but also impractical in large or crowded environments.

This project, "Face Mask Detection", aims to develop an intelligent system capable of identifying whether a person is wearing a face mask or not. The system is trained using an image dataset containing labeled images of individuals with and without face masks. By learning from this dataset, the model can recognize visual differences between masked and unmasked faces.

The core of the project relies on techniques that classify images based on facial features. The main steps involved in this project are preprocessing, feature extraction, and classification. In preprocessing, the images are cleaned and prepared. Then, feature extraction is the process of identifying and collecting important parts or patterns from an image. Finally, in classification, the system decides whether the person in the image is wearing a mask or not.

In real-world applications, face mask detection systems can be integrated into surveillance cameras, entry gates, and mobile devices to ensure compliance in places such as airports, hospitals, schools, and shopping malls. These systems can operate in real time, providing instant alerts when individuals are detected without masks. This not only enhances public safety but also helps authorities enforce health regulations more effectively. Moreover, the scalability of such systems allows them to be deployed across multiple locations simultaneously, making them a valuable asset in managing large crowds and high-traffic areas.

To ensure reliability and adaptability, the system can be trained on diverse datasets that include variations in lighting, angles, and facial expressions. This helps maintain consistent performance across different environments and user demographics. Additionally, the system can be updated periodically with new data to improve its accuracy and responsiveness. By automating the process of mask detection, this project contributes to building safer public spaces and demonstrates the potential of intelligent systems in addressing global health challenges.

1.2 MOTIVATION

The rapid spread of airborne diseases like COVID-19 has underscored the need for proactive and scalable health monitoring solutions. In environments such as transportation hubs, educational institutions, and workplaces, enforcing mask-wearing policies is essential to minimize transmission risks. Yet, relying solely on human supervision is not only inefficient but also exposes staff to potential health hazards. This challenge calls for a technological intervention that can operate autonomously and consistently.

Face mask detection systems offer a promising approach to bridge this gap. By automating the process of identifying mask compliance, these systems reduce human error and ensure uniform enforcement of safety protocols. They can be deployed in real-time surveillance setups, helping authorities respond swiftly to violations and maintain a safer public environment. The motivation behind this research is to harness the power of visual recognition to create a tool that is not only accurate and fast but also adaptable to various settings and populations.

Moreover, the increasing accessibility of digital cameras and smart devices makes it feasible to implement such systems on a wide scale. Whether integrated into mobile apps, kiosks, or CCTV networks, face mask detection can become a standard feature in public health infrastructure. This research is driven by the vision of building intelligent tools that support long-

term health resilience, not just during pandemics but also in preparation for future outbreaks. It reflects a commitment to innovation that serves both individual safety and collective well-being.

1.3 OBJECTIVE

The primary objective of this research is to design and implement a robust system that can automatically detect whether individuals are wearing face masks in real-time visual environments. This system aims to support public health efforts by enabling contactless monitoring of mask compliance in various public and private settings.

A key goal is to ensure that the detection system performs accurately across diverse conditions, including different lighting environments, facial orientations, and mask types. The system should be capable of distinguishing between properly worn masks, improperly worn masks, and the absence of masks, thereby offering nuanced insights into compliance levels.

Another important objective is to create a solution that is scalable and adaptable for integration into existing infrastructure such as surveillance cameras, mobile applications, and access control systems. The system should be lightweight enough to operate efficiently on edge devices while maintaining high detection precision.

This research also seeks to contribute to the broader field of intelligent safety systems by demonstrating how visual recognition technologies can be leveraged to solve real-world problems. By focusing on practical deployment and user-friendly design, the project aims to make face mask detection accessible to institutions ranging from healthcare facilities to educational campuses.

Ultimately, the objective is not only to build a functional tool but also to promote a culture of safety and responsibility through technology. The system should serve as a preventive measure

that complements human efforts, reduces manual workload, and enhances the overall effectiveness of public health protocols.

1.4 OVERVIEW OF WORKING PROCESS

The face mask detection system operates through a structured pipeline that transforms raw image inputs into actionable insights about mask compliance. The process begins with the acquisition of a diverse set of facial images, including individuals wearing masks and those without. These images are carefully labeled to guide the supervised learning process. The dataset is designed to reflect real-world variability in lighting, facial orientation, and mask types, ensuring that the system can generalize effectively across different environments.

Once the labeled data is prepared, it is used to train deep learning models capable of recognizing visual patterns associated with mask usage. Convolutional Neural Networks (CNNs) are employed for their ability to learn spatial hierarchies and detect subtle differences in facial regions. Additionally, MobileNetV2 is integrated for its lightweight architecture and fast inference capabilities, making the system suitable for deployment on mobile and embedded platforms. The training process involves multiple iterations, during which the model learns to associate image inputs with correct mask status labels.

After training, the model is deployed into a real-time detection framework. This setup allows the system to analyze live video feeds or static images and classify individuals as either “Mask Worn” or “No Mask.” The classification results can be used to trigger alerts, log compliance data, or integrate with access control systems. The system is designed to operate efficiently, delivering rapid and reliable predictions even in crowded or fast-moving environments.

To ensure reliability, the model undergoes rigorous evaluation using metrics such as accuracy, precision, recall, and F1-score. A separate test dataset is used to validate the model's performance and identify areas for improvement. The system's modular design also supports future enhancements, such as detecting improper mask usage or integrating with thermal screening tools. Overall, the working process reflects a blend of intelligent design and practical deployment, aimed at supporting public health through automated, contactless monitoring.

1.5 IMPORTANCE OF FACE MASK DETECTION

Face mask detection has emerged as a vital technological solution in the effort to control the spread of airborne diseases, particularly during pandemics like COVID-19. Wearing a face mask significantly reduces the transmission of respiratory droplets, which are the primary carriers of viruses. However, ensuring that individuals consistently follow mask-wearing guidelines in public spaces is a major challenge, especially in crowded environments. Automated face mask detection systems help bridge this gap by providing a contactless, efficient way to monitor compliance without relying on manual enforcement.

The significance of such systems extends beyond health safety—they also contribute to operational efficiency. In places like airports, hospitals, schools, and shopping centers, deploying face mask detection technology can streamline entry protocols and reduce the need for human personnel to check each individual. This not only minimizes exposure risks for staff but also speeds up the flow of people through checkpoints, making public spaces safer and more manageable.

Moreover, face mask detection supports data-driven decision-making. By analyzing patterns of compliance across different locations and times, authorities can identify hotspots of non-compliance and take targeted action. This proactive approach helps in resource allocation,

policy enforcement, and public awareness campaigns. It also fosters a sense of accountability among citizens, knowing that their actions are being monitored in a non-intrusive and fair manner.

From a technological standpoint, integrating face mask detection into surveillance systems, mobile applications, and smart devices demonstrates the power of artificial intelligence in solving real-world problems. It showcases how deep learning models like CNN and MobileNetV2 can be trained to recognize subtle visual cues and make accurate predictions in real time. These models enable the system to function reliably across diverse conditions, including varying lighting, angles, and mask types.

In essence, face mask detection is not just a tool for pandemic response—it is a forward-looking solution that strengthens public health infrastructure. As societies prepare for future health emergencies, such systems will play a crucial role in maintaining safety, building trust, and promoting responsible behavior. Their importance lies in their ability to combine technology with public service, creating smarter environments that prioritize well-being without compromising efficiency.

1.6 ROLE OF DEEP LEARNING IN SOLVING THESE CHALLENGES

Deep learning has revolutionized the way visual recognition tasks are approached, making it a powerful tool for face mask detection. Unlike traditional methods that rely on manual feature selection, deep learning models can automatically learn and extract meaningful patterns from raw image data. This capability is especially important when dealing with diverse facial appearances, lighting conditions, and mask styles, which are common in real-world scenarios.

Convolutional Neural Networks (CNNs) are particularly effective in this domain due to their ability to capture spatial hierarchies and fine-grained details in images. They help the system

distinguish between masked and unmasked faces by focusing on critical facial regions such as the nose and mouth. CNNs also support high accuracy and adaptability, making them suitable for dynamic environments where facial orientations and expressions vary.

MobileNetV2 further enhances the system's practicality by offering a lightweight architecture optimized for mobile and embedded devices. It delivers fast inference speeds with minimal computational resources, allowing the detection system to run efficiently on smartphones, surveillance cameras, and other edge devices. This makes real-time monitoring feasible even in resource-constrained settings. deep learning enables scalable, accurate, and responsive face mask detection. Its ability to learn from data, adapt to new conditions, and operate in real time makes it an essential component in building intelligent safety systems that support public health efforts.

1.7 SCOPE

The scope of this project encompasses the development of an intelligent, image-based system capable of detecting whether individuals are wearing face masks in real-time or from static images. It focuses on building a reliable and scalable solution that can be integrated into various platforms such as surveillance systems, mobile applications, and access control devices. The system is designed to function effectively across diverse environments, accounting for variations in lighting, facial orientation, and mask types. It includes stages such as data preprocessing, feature extraction, and classification using deep learning models like CNN and MobileNetV2 to ensure high accuracy and efficiency.

This project is limited to visual detection and does not extend to identifying individuals or verifying mask quality. It assumes that the input images or video frames are of sufficient resolution to detect facial features clearly. The system is intended for public health monitoring and not for law enforcement or biometric identification. While the current focus is on binary classification—

mask or no mask—it can be extended in future work to include improper mask usage or integration with thermal screening for enhanced safety. The scope also includes performance evaluation and testing to ensure the model's robustness in real-world deployment scenarios.

CHAPTER-2

2.1 LITERATURE SURVEY

The outbreak of the COVID-19 pandemic has had an unprecedented impact on global public health systems, societal behavior, and technological innovation. One of the most immediate and widely adopted interventions has been the use of face masks, which serve as a primary barrier to the transmission of SARS-CoV-2, the virus responsible for COVID-19 [1]. Scientific studies and recommendations from organizations like the World Health Organization (WHO) have repeatedly confirmed that consistent and proper use of masks significantly reduces the spread of respiratory droplets, which are the main carriers of viral particles. The rapid and extensive transmission of the virus prompted governments worldwide to implement mandatory mask-wearing policies in public spaces such as hospitals, educational institutions, public transportation, shopping areas, and crowded urban centers. These regulations, while effective, also introduced the challenge of ensuring compliance across large populations, particularly in areas with dense foot traffic and limited surveillance personnel.

Manual monitoring of mask compliance is labor-intensive and prone to errors, motivating the adoption of intelligent systems powered by Artificial Intelligence (AI) and Machine Learning (ML) [2][3]. Computer vision algorithms, particularly those leveraging deep learning, can detect whether individuals are wearing masks in real-time, track non-compliance trends, and issue alerts to administrators when violations exceed a predefined threshold. These systems utilize convolutional neural networks (CNNs) to extract high-dimensional facial features, even under partial occlusion caused by masks, glasses, or head coverings. Hybrid approaches that combine deep learning for feature extraction with traditional machine learning classifiers such as Support Vector Machines (SVM) or Random Forests (RF) have demonstrated remarkable

accuracy and computational efficiency [6][7]. Such systems are capable of detecting multiple faces simultaneously in crowded environments, adjusting to variations in lighting, orientation, and distance, thereby overcoming many limitations associated with traditional face detection techniques.

The development of robust face mask detection models relies heavily on high-quality datasets. Various datasets, including RMFD, SMFD, MFDD, and RMFRD, have been curated to encompass a wide range of demographics, facial orientations, mask types, and occlusion scenarios [17]. These datasets provide essential training and validation material for deep learning models, enabling them to generalize effectively across real-world environments. Advanced data augmentation strategies, such as synthetic mask application and geometric transformations, are often employed to further enhance model robustness and prevent overfitting. Additionally, the integration of Generative Adversarial Networks (GANs) has enabled the reconstruction of occluded facial features, facilitating accurate recognition of masked individuals while maintaining privacy considerations [22][23]. These technological interventions allow authorities to monitor mask usage efficiently, analyze compliance trends, and implement targeted interventions for high-risk areas.

The role of AI in pandemic mitigation extends beyond mere surveillance. By automating mask detection, these systems reduce the need for direct human oversight, thereby minimizing exposure risk for personnel while improving the speed and accuracy of enforcement. Furthermore, the deployment of AI-enabled mask detection supports epidemiological studies by providing granular data on mask compliance across different regions and population groups. Behavioral studies indicate that individuals wear masks not only to prevent viral infection but also to reduce exposure to air pollution, to comply with public health regulations, or to maintain

privacy in social settings [2]. Consequently, automated systems must be designed to handle a variety of contexts and behaviors without infringing on personal freedoms or privacy.

Technological evolution in this domain has been rapid. Early methods relied on traditional feature-based algorithms, while modern approaches utilize deep neural networks and object detection frameworks such as YOLO, Faster R-CNN, and RetinaFace [26][27]. These models have been trained to detect subtle variations in facial structures, distinguish masked from unmasked faces, and recognize faces in dynamic and complex environments. Incorporating attention mechanisms, transfer learning, and multi-granularity features has further enhanced detection accuracy, achieving rates as high as 99% in controlled experiments [16][17]. The integration of these methods into real-time camera systems, mobile devices, and edge computing platforms allows continuous monitoring of public spaces without significant infrastructure costs.

The application of AI-powered mask detection is not limited to compliance monitoring; it also contributes to public safety and pandemic preparedness. In healthcare settings, for example, automated detection can ensure that staff, patients, and visitors adhere to safety protocols, reducing the risk of nosocomial infections. In educational and corporate environments, these systems can maintain safe operational conditions by alerting administrators to potential violations. Moreover, AI-driven analytics can provide insights into patterns of mask-wearing behavior, informing public health campaigns and targeted interventions [4][17][21]. By combining surveillance, predictive analytics, and automated enforcement, these intelligent systems create a multi-layered defense against viral transmission.

Despite the promising performance of AI-based mask detection systems, challenges remain. Variations in lighting, camera quality, face orientation, and mask type can introduce errors in detection. Occlusions caused by scarves, glasses, or multiple overlapping masks can also reduce model accuracy. To address these issues, researchers are exploring hybrid architectures that combine classical image processing techniques with deep learning, as well as multi-stage detection pipelines that incorporate face alignment, feature enhancement, and contextual reasoning [2][6][7]. Future advancements are likely to include the integration of multimodal data sources, such as thermal imaging and depth sensors, which can further improve detection reliability in complex scenarios.

In summary, the convergence of AI, ML, and computer vision has created a powerful framework for face mask detection, which is crucial for public health management during the COVID-19 pandemic and future respiratory disease outbreaks. By providing real-time, accurate, and automated monitoring, these systems not only ensure compliance with mask regulations but also enhance public safety, reduce exposure risk for personnel, and generate valuable data for epidemiological analysis. The ongoing evolution of these technologies promises even greater robustness, efficiency, and applicability across diverse environments, marking a significant milestone in the intersection of artificial intelligence and global health management [1][2][17][21][26][27].

The evolution of computer vision for face mask detection has been driven primarily by the rapid advancements in deep learning architectures. Traditional face recognition systems struggled significantly with partial occlusion, where masks hide essential facial landmarks such as the nose, lips, and chin. Earlier approaches based on hand-crafted features—such as Haar cascades, Histogram of Oriented Gradients (HOG), and Local Binary Patterns (LBP)—were unable to generalize across variations in mask designs, lighting conditions, or ethnic facial

structures [26]. In contrast, convolutional neural networks (CNNs) automatically learn hierarchical features that capture fine-grained visual patterns and spatial relationships, making them ideal for identifying masked and unmasked faces in diverse real-world settings.

A major breakthrough in this domain has been the adaptation of object detection algorithms like Faster R-CNN, YOLO (You Only Look Once), and SSD (Single Shot Multibox Detector) for face mask recognition. These models operate by dividing an image into regions, predicting bounding boxes, and classifying each region as either “mask” or “no mask.” The Faster R-CNN model, in particular, employs a Region Proposal Network (RPN) to identify potential object regions before classification, achieving a high balance between accuracy and inference time [27]. Similarly, YOLOv5 and YOLOv7 models perform detection in a single forward pass, allowing real-time monitoring in high-traffic environments such as airports, hospitals, and shopping malls [30]. These models have been optimized through transfer learning, where pre-trained weights from large datasets such as COCO or ImageNet are fine-tuned on mask detection datasets. This approach not only accelerates training but also enhances performance in limited-data scenarios.

In addition to model selection, dataset quality and preprocessing play a crucial role in the success of face mask detection. Datasets such as RMFD (Real-World Masked Face Dataset), SMFD (Simulated Masked Face Dataset), and MAFA (Masked Faces) are widely used to train deep models to recognize masked individuals across variations in pose, lighting, and background clutter. Data augmentation techniques—including random cropping, flipping, rotation, scaling, and synthetic mask overlay—expand the dataset’s diversity, enabling models to generalize better to unseen data. Some researchers have also utilized Generative Adversarial Networks (GANs) to synthesize realistic masked faces, thereby enhancing dataset richness without additional data collection costs [31]. GAN-based synthetic augmentation helps

simulate edge cases, such as partially worn masks, transparent masks, or cultural face coverings, improving the model's robustness under challenging scenarios.

Furthermore, hybrid systems that combine deep learning with classical image processing have emerged as a powerful solution for edge-based applications. For example, lightweight CNNs like MobileNetV2, EfficientNet, and SqueezeNet have been deployed on embedded systems, drones, and IoT cameras for real-time detection with low computational cost [24][30]. These networks maintain high accuracy while minimizing memory usage, making them suitable for deployment on low-power devices. Coupled with cloud-based analytics, such systems can aggregate compliance data from multiple cameras, generate statistical dashboards, and trigger automated alerts when non-compliance exceeds certain thresholds.

One significant research direction focuses on multi-stage detection frameworks. In such frameworks, the first stage performs face localization, while the second stage determines mask compliance. Some models, like those based on Mask R-CNN, extend beyond detection by segmenting the mask region at the pixel level, enabling fine-grained analysis of mask fit and coverage [13][14]. This approach is particularly valuable in healthcare settings where partial mask wearing (below the nose, for instance) can still pose significant health risks. By integrating semantic segmentation and classification, Mask R-CNN-based methods offer both qualitative and quantitative assessments of mask usage, ensuring more reliable enforcement of safety standards.

To improve real-time adaptability, researchers have also introduced attention mechanisms and multi-scale feature extraction modules into CNN architectures. These components allow the model to focus dynamically on relevant facial regions while ignoring irrelevant background noise. The integration of Feature Pyramid Networks (FPN) further enhances the model's ability

to detect faces across varying scales, from close-up individuals to distant groups [25][28]. Recent innovations also involve context-aware detection, where environmental cues (such as proximity between individuals or crowd density) are incorporated into the system's decision-making logic, enabling proactive alerts in high-risk scenarios.

From an application standpoint, AI-powered mask detection systems have been widely deployed across several domains. In transportation hubs, such as airports and railway stations, they assist security staff in enforcing health protocols efficiently. In smart cities, these systems are integrated with surveillance networks to track compliance trends over time. In educational institutions, they ensure students adhere to safety regulations while minimizing disruptions to daily operations. Moreover, the healthcare sector has benefited immensely from automated mask detection, as it provides an additional layer of protection for frontline workers by ensuring that all entrants comply with safety requirements before entering critical zones such as ICUs or operating rooms [6][12].

Despite these successes, several challenges persist. False positives and false negatives may occur due to poor lighting, low-resolution images, or overlapping faces in crowded environments. Environmental conditions such as rain, fog, or motion blur can further degrade detection accuracy. Moreover, variations in mask texture, transparency, and design (such as patterned or colored masks) may cause misclassification. Researchers have begun addressing these issues using ensemble learning, where multiple models are combined to make a collective decision, thus improving reliability under uncertain conditions. In addition, the use of edge AI and federated learning is gaining attention for privacy-preserving applications. These techniques ensure that sensitive facial data remain on local devices, aligning with global data protection regulations such as GDPR.

As the world gradually transitions to post-pandemic normalcy, the utility of mask detection extends beyond COVID-19. These systems can serve as the foundation for broader public health monitoring frameworks, supporting disease surveillance, crowd management, and even emotional recognition while preserving individual privacy. Integration with biometric security systems can create multi-functional platforms capable of verifying identity and health compliance simultaneously. Moreover, the lessons learned from AI-based mask detection can inform the development of future technologies aimed at combating airborne diseases, enhancing workplace safety, and ensuring preparedness for potential pandemics [21][23][31].

In conclusion, the convergence of computer vision, deep learning, and AI-driven analytics has fundamentally transformed the way mask detection and public health monitoring are conducted. From simple classification models to sophisticated multi-stage and context-aware frameworks, the field has evolved to meet the complex demands of real-world implementation. With continued research into explainable AI, federated learning, and lightweight architectures, the next generation of face mask detection systems will offer greater transparency, speed, and adaptability. These technologies not only represent a triumph of innovation in response to a global crisis but also lay the groundwork for smarter, safer, and more resilient societies.

CHAPTER-3

3.1 DATASET DESCRIPTION

3.1.1 Data Collection

The dataset used in this study is sourced from Kaggle and titled *Face Mask Detection Dataset* by Omkar Gurav. It contains a curated set of facial images categorized based on mask usage. The dataset plays a foundational role in training and validating the deep learning model for accurate classification of masked and unmasked faces.

The images are collected under varied conditions to simulate real-world scenarios. These include differences in lighting, facial orientation, background clutter, and mask styles. The dataset is designed to support supervised learning, where each image is labeled to indicate whether the subject is wearing a mask or not.

To ensure model generalization, the dataset includes individuals of different age groups, genders, and ethnicities. This diversity helps the model learn robust features and perform reliably across different environments. The dataset is split into training, validation, and test sets to support model development and performance evaluation.

3.1.2 Purpose

The primary purpose of collecting this dataset is to enable the development of an automated system that can detect face mask usage in real time. With increasing public health concerns, especially during pandemics, monitoring mask compliance has become essential. Manual enforcement is labor-intensive and risky, making automated solutions highly desirable.

This dataset provides the necessary labeled visual data to train deep learning models capable of distinguishing between masked and unmasked faces. It supports the creation of a scalable

and deployable system that can be used in public spaces such as hospitals, schools, and transportation hubs.

Additionally, the dataset serves academic and research purposes by offering a benchmark for evaluating different model architectures. It allows researchers to experiment with various preprocessing techniques, model configurations, and performance metrics.

3.1.3 Key Characteristics

Summary of the key characteristics of the dataset:

Attribute	**Details**
Total Images	7,553
Categories	With Mask, Without Mask
Image Format	JPEG / PNG
Resolution	Varies (typically 512×512 or less)
Source	Kaggle Dataset

3.1.4 Sample Dataset Image

To provide visual context for the dataset used in this study, a sample image from the *Face Mask Detection Dataset* is included below. This image illustrates the type of data used to train and evaluate the deep learning model for mask classification.

The dataset comprises facial images categorized into two classes: **With Mask** and **Without Mask**. Each image is labeled accordingly and reflects real-world conditions such as varied lighting, facial orientations, and mask styles. The sample image shown here belongs to the "With Mask" category and demonstrates the clarity and resolution typical of the dataset.

Including sample images in the documentation helps convey the nature and quality of the data. It also supports reproducibility by allowing other researchers to understand the visual characteristics that the model was trained on. The presence of consistent labeling and diverse facial features ensures that the model can generalize effectively across different populations.

This visual reference complements the tabular summary provided earlier and reinforces the importance of using well-annotated, representative data in supervised learning tasks. It serves as a foundation for the model's ability to detect mask usage accurately and reliably.

CHAPTER-4

METHODOLOGY

4.1 Introduction

The methodology section outlines the systematic approach adopted to design, develop, and evaluate the face mask detection system. This project leverages deep learning techniques to automate the identification of face mask usage in real-time environments. The methodology is structured to ensure that each phase—from data acquisition to deployment—is logically connected and technically sound.

The primary goal is to build a robust and scalable system that can operate efficiently across diverse conditions such as varying lighting, facial orientations, and mask styles. The methodology emphasizes the use of lightweight yet powerful deep learning models to ensure real-time performance without compromising accuracy.

This section also highlights the modular design of the system, which allows for easy integration into existing surveillance or access control infrastructures. Each module is developed with flexibility in mind, enabling future enhancements such as improper mask detection or integration with thermal screening systems.

4.2 Methodology

The methodology follows a supervised learning framework, where a labeled dataset of facial images is used to train a deep learning model. The process is divided into several key stages: data collection, model training, classification, evaluation, and system modularization. Each stage is carefully designed to contribute to the overall performance and reliability of the system.

Deep learning models, particularly Convolutional Neural Networks (CNNs) and MobileNetV2, are employed for their ability to learn complex visual features. These models are trained to

distinguish between masked and unmasked faces using spatial hierarchies and pattern recognition. The choice of MobileNetV2 ensures that the system remains lightweight and suitable for deployment on edge devices.

The methodology also incorporates real-time detection capabilities, allowing the system to process live video feeds and generate immediate alerts. This is crucial for applications in public spaces where timely intervention is necessary. The system is evaluated using standard performance metrics to ensure its effectiveness in real-world scenarios.

4.3 Data Collection

Data collection is a foundational step that directly impacts the model's learning and generalization capabilities. A diverse dataset is curated, containing images of individuals with and without face masks. The dataset includes variations in age, gender, ethnicity, lighting conditions, facial expressions, and mask types to ensure that the model can perform reliably across different populations.

Publicly available datasets such as RMFD (Real-World Masked Face Dataset) and MAFA (Masked Faces) are used, along with custom images captured in controlled environments. Each image is manually labeled to indicate the presence or absence of a mask, forming the basis for supervised learning. Care is taken to balance the dataset to prevent bias toward any particular class.

To enhance the dataset's robustness, data augmentation techniques such as rotation, flipping, and brightness adjustment are applied. This increases the variety of training samples and helps the model generalize better to unseen data. The final dataset is split into training, validation, and test sets to support model development and evaluation.

4.4 Model Training

Model training involves feeding the labeled image data into deep learning architectures to learn the visual patterns associated with mask usage. The training process is conducted using CNNs and MobileNetV2, both of which are well-suited for image classification tasks. CNNs are used for their ability to extract hierarchical features, while MobileNetV2 is chosen for its efficiency and suitability for mobile deployment.

The dataset is divided into training and validation sets, typically in an 80:20 ratio. The model is trained over multiple epochs using optimization algorithms such as Adam or Stochastic Gradient Descent (SGD). Loss functions like categorical cross-entropy are used to measure prediction errors, and accuracy is tracked as a primary performance metric.

Regularization techniques such as dropout and early stopping are employed to prevent overfitting and improve generalization. Batch normalization is also used to stabilize and accelerate training. The model's performance is monitored throughout the training process, and hyperparameters are tuned to achieve optimal results.

4.5 Classification

Once the model is trained, it is used to classify new input images into predefined categories: "Mask Worn" or "No Mask." The classification process involves passing an image through the trained model, which outputs a probability score for each class. The class with the highest score is selected as the final prediction.

The system is designed to perform classification in real time, making it suitable for live surveillance and access control applications. It can process video frames continuously and provide instant feedback on mask compliance. This enables timely interventions and enhances public safety in high-traffic areas.

The classification module is also designed to be extensible. With additional training data, it can be adapted to detect improper mask usage, such as masks worn below the nose or loosely fitted. This flexibility allows the system to evolve with changing public health guidelines and user requirements.

4.5.1 Convolutional Neural Networks (CNN)

Convolutional Neural Networks (CNNs) are a specialized class of deep learning models that have revolutionized the field of computer vision. Unlike traditional neural networks, CNNs are designed to automatically and adaptively learn spatial hierarchies of features from images. This makes them particularly effective for tasks such as image classification, object detection, and facial recognition.

Architecture and Working Principle

The core idea behind CNNs is the use of convolutional layers that apply filters (also called kernels) to input images. These filters slide across the image, performing element-wise multiplication and summation to produce feature maps. Each filter is capable of detecting specific patterns such as edges, corners, or textures. As the network goes deeper, it learns increasingly abstract features, enabling it to recognize complex shapes and objects.

A typical CNN architecture includes:

- **Convolutional Layers**: Extract features from the input image.
- **Activation Functions (e.g., ReLU)**: Introduce non-linearity to help the network learn complex patterns.
- **Pooling Layers (e.g., Max Pooling)**: Reduce the spatial dimensions of feature maps, retaining the most important information while reducing computational load.

- **Fully Connected Layers**: Flatten the feature maps and perform classification based on learned features.

Advantages of CNNs

CNNs offer several advantages over traditional machine learning models:

- **Automatic Feature Extraction**: They eliminate the need for manual feature engineering.
- **Translation Invariance**: Pooling layers help the model recognize objects regardless of their position in the image.
- **Parameter Efficiency**: Shared weights in convolutional layers reduce the number of parameters, making training more efficient.

Application in Face Mask Detection

In the context of face mask detection, CNNs are used to analyze facial images and determine whether a mask is present. The model learns to focus on key facial regions such as the nose, mouth, and jawline—areas typically covered by a mask. By training on a labeled dataset of masked and unmasked faces, the CNN can generalize across different lighting conditions, facial orientations, and mask styles.

CNNs are particularly useful in scenarios where high accuracy is required and computational resources are available. They form the backbone of many state-of-the-art face mask detection systems and can be further enhanced by integrating with lightweight models for real-time deployment.

4.5.2 MobileNetV2

MobileNetV2 is a deep learning architecture developed by Google, specifically designed for mobile and embedded vision applications. It builds upon the original MobileNet by introducing architectural innovations that improve both performance and efficiency. MobileNetV2 is widely used in real-time systems where computational resources are limited but accuracy cannot be compromised.

Key Innovations

MobileNetV2 introduces two major concepts:

- **Depthwise Separable Convolutions**: These break down standard convolutions into two operations—depthwise and pointwise. Depthwise convolutions apply a single filter per input channel, while pointwise convolutions (1x1 convolutions) combine the outputs. This drastically reduces the number of parameters and computations.
- **Inverted Residuals with Linear Bottlenecks**: Unlike traditional residual blocks, MobileNetV2 expands the input channels before applying depthwise convolutions and then compresses them back using linear bottlenecks. This structure allows for efficient feature reuse and better gradient flow during training.

Advantages of MobileNetV2

MobileNetV2 offers several benefits that make it ideal for real-time applications:

- **Lightweight Architecture**: It has fewer parameters and lower memory requirements compared to models like ResNet or VGG.
- **Fast Inference**: Suitable for deployment on devices with limited processing power, such as smartphones, Raspberry Pi, or surveillance cameras.

- **Competitive Accuracy**: Despite its compact size, MobileNetV2 achieves performance comparable to larger models on many image classification tasks.

Application in Face Mask Detection

MobileNetV2 is particularly well-suited for face mask detection in real-time environments. Its lightweight design allows it to process video frames quickly, making it ideal for continuous monitoring in public spaces. The model can be trained on a dataset of facial images and deployed on edge devices to detect mask compliance without requiring cloud-based computation.

In practice, MobileNetV2 can be integrated into surveillance systems, mobile apps, or smart kiosks. It enables fast and accurate detection of masked and unmasked faces, supporting automated alerts and compliance logging. Its efficiency ensures that the system remains responsive even under heavy load, such as monitoring large crowds.

Comparative Insights and Integration

While CNNs provide a powerful framework for feature extraction and classification, MobileNetV2 offers a streamlined version optimized for speed and resource efficiency. In many face mask detection systems, CNNs are used during the initial development and training phases, while MobileNetV2 is employed for deployment and real-time inference.

The integration of both models allows researchers and developers to balance accuracy and performance. For example, a CNN can be used to train a high-accuracy model on a large dataset, and then the learned weights can be transferred to a MobileNetV2 architecture for deployment. This approach combines the strengths of both models and ensures that the system is both robust and scalable.

4.6 Model Evaluation

Model evaluation is conducted to assess the system's accuracy, reliability, and robustness. A separate test dataset, not used during training, is employed to evaluate the model's generalization ability. Key performance metrics include accuracy, precision, recall, and F1-score, which provide a comprehensive view of the model's strengths and weaknesses.

A confusion matrix is generated to visualize the model's classification performance. It highlights true positives, false positives, true negatives, and false negatives, helping identify areas where the model may need improvement. This analysis is crucial for refining the model and ensuring its readiness for deployment.

The evaluation process also includes stress testing under varied conditions, such as low lighting, partial occlusion, and different camera angles. This ensures that the model performs reliably in real-world scenarios. Based on the evaluation results, the model may be retrained or fine-tuned to enhance its performance.

4.7 Module Description

The system is composed of several functional modules, each responsible for a specific task in the detection pipeline. These modules work together to deliver a seamless and efficient face mask detection experience. The modular design allows for easy maintenance, upgrades, and integration into existing infrastructures.

- **Input Module**: Captures images or video frames from cameras or uploaded sources. It serves as the entry point for data into the system.
- **Model Interface Module**: Handles the interaction between the input data and the trained model. It ensures that the data is correctly formatted and passed to the model for prediction.

- **Classification Module**: Uses the trained model to determine mask status and outputs the result along with a confidence score.
- **Alert Module**: Generates notifications, logs, or triggers based on detection outcomes. It can be integrated with external systems for automated responses.

Each module is designed to be independent yet interoperable, allowing for future enhancements such as multi-class detection, integration with thermal cameras, or cloud-based analytics.

4.8 Significance of the Study

The significance of this study lies in its contribution to public health and safety through the application of intelligent systems. In the wake of global health crises such as the COVID-19 pandemic, ensuring compliance with preventive measures like face mask usage has become critical. Manual enforcement is not only labor-intensive but also exposes personnel to unnecessary risk. This research offers a technological solution that automates mask detection, reducing human intervention and enhancing safety in public spaces.

From a technical perspective, the study demonstrates how deep learning models—particularly CNN and MobileNetV2—can be effectively applied to real-time image classification tasks. It showcases the potential of lightweight architectures in delivering high accuracy while maintaining computational efficiency, making the system suitable for deployment on mobile and embedded devices. This is especially important for resource-constrained environments such as schools, clinics, and rural health centers.

Academically, the study contributes to the growing field of computer vision applications in healthcare. It provides a replicable methodology for researchers interested in developing similar systems for other forms of visual compliance monitoring, such as hand hygiene detection or

social distancing analysis. The modular design of the system also supports future enhancements, allowing researchers to expand its capabilities based on evolving public health needs.

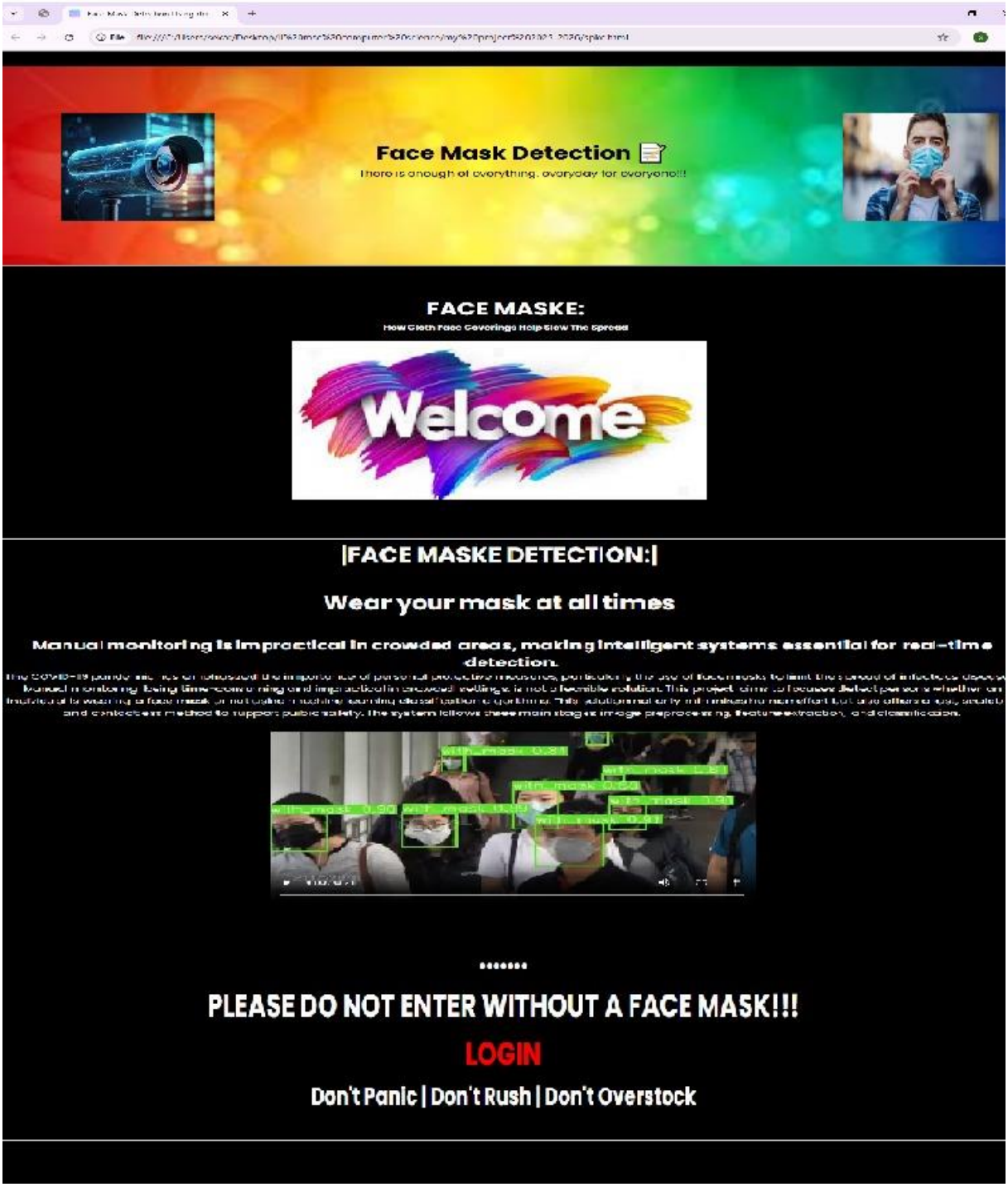
Socially, the system promotes responsible behavior by providing a non-intrusive way to monitor mask usage. It encourages individuals to adhere to safety protocols and supports institutions in maintaining a secure environment. By bridging the gap between policy enforcement and technological innovation, this study plays a vital role in shaping smarter, safer communities.

4.9 Limitations

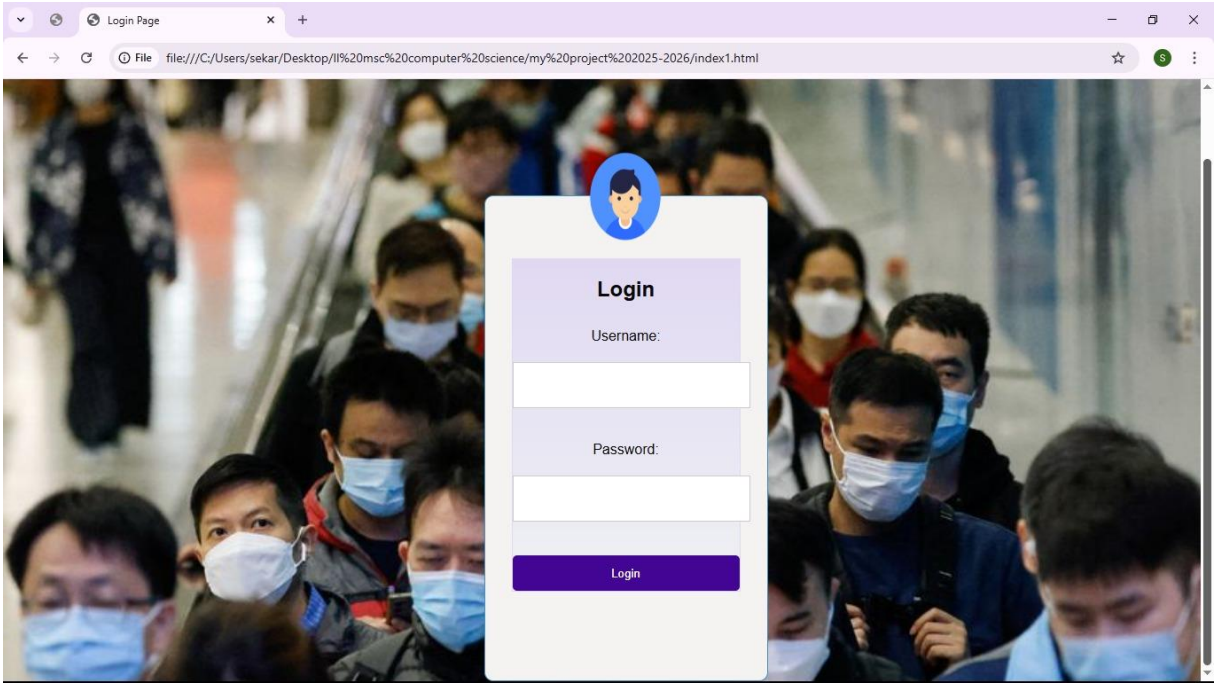
While the system performs well under most conditions, it has certain limitations. It may struggle with extreme lighting variations, heavily occluded faces, or very low-resolution images. Additionally, the current model focuses on binary classification and does not yet detect improper mask usage.

Future enhancements include expanding the classification categories to include “Improper Mask,” integrating thermal screening for fever detection, and adding multilingual voice alerts. The system can also be extended to support cloud-based analytics for centralized monitoring across multiple locations.

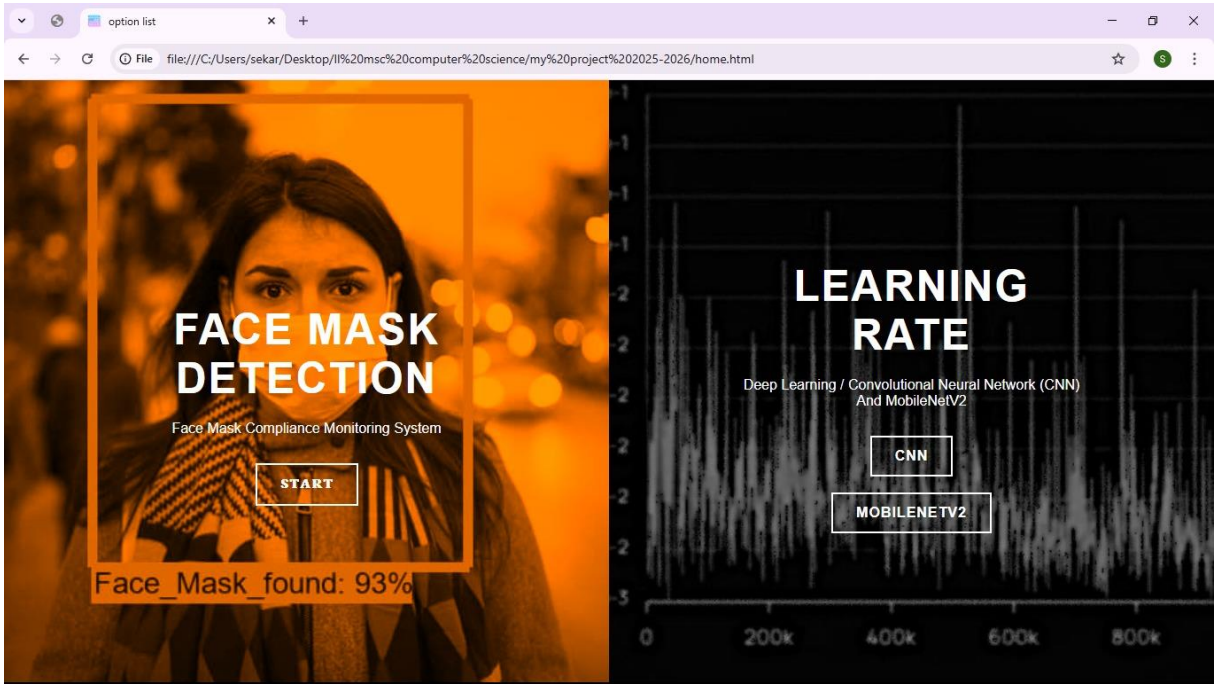
HOME PAGE


Face Mask Detection
There is enough of everything, everyday for everyone!!!
FACE MASKE:
How Cloth Face Coverings Help Slow The Spread
Welcome
|FACE MASKE DETECTION:|
Wear your mask at all times
Manual monitoring is impractical in crowded areas, making intelligent systems essential for real-time detection.
PLEASE DO NOT ENTER WITHOUT A FACE MASK!!!
LOGIN
Don't Panic | Don't Rush | Don't Overstock

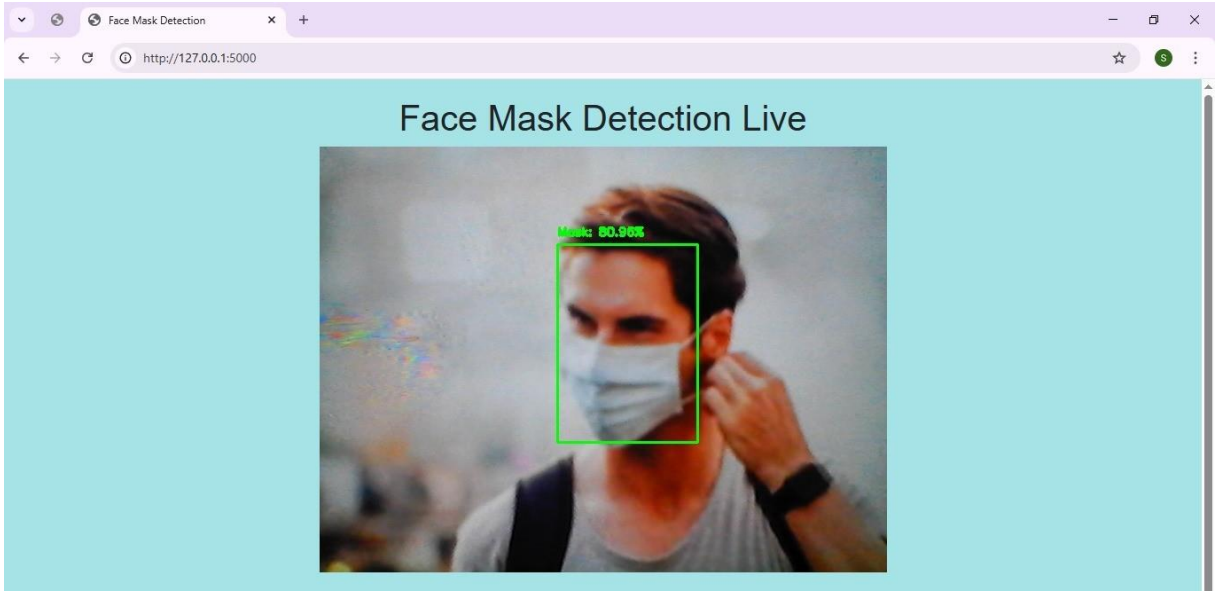
LOGIN FORM



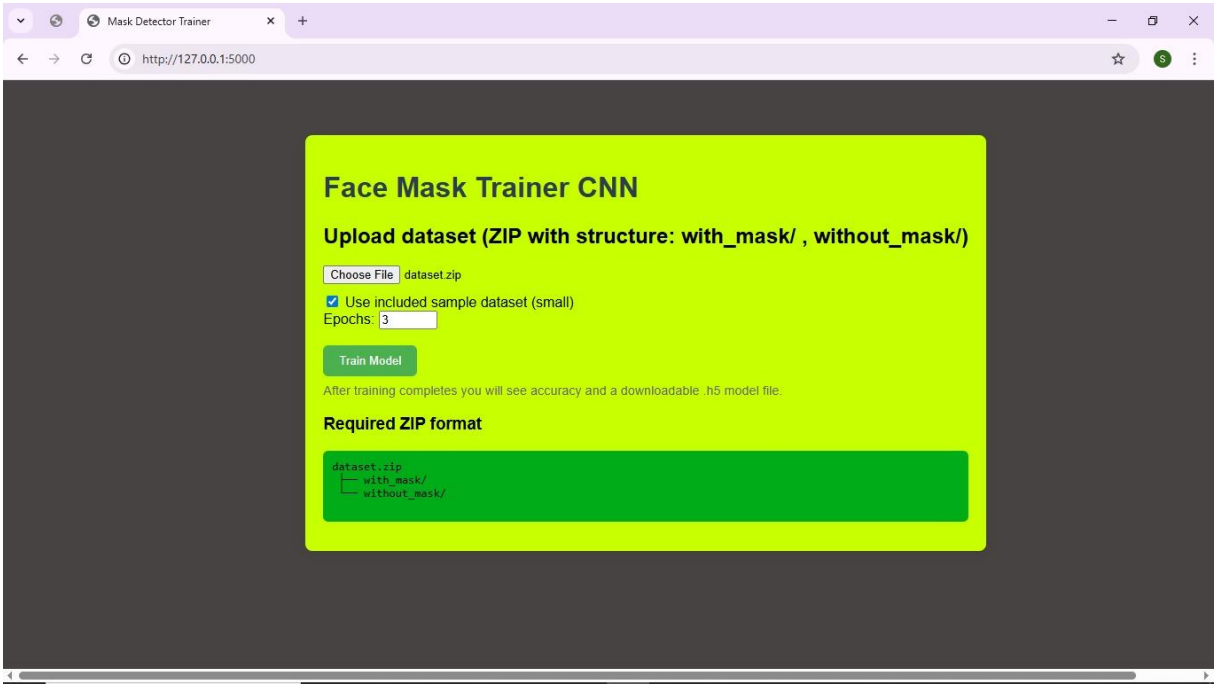
FACE MASK DETECTION & LEARNING RATE HOME PAGE



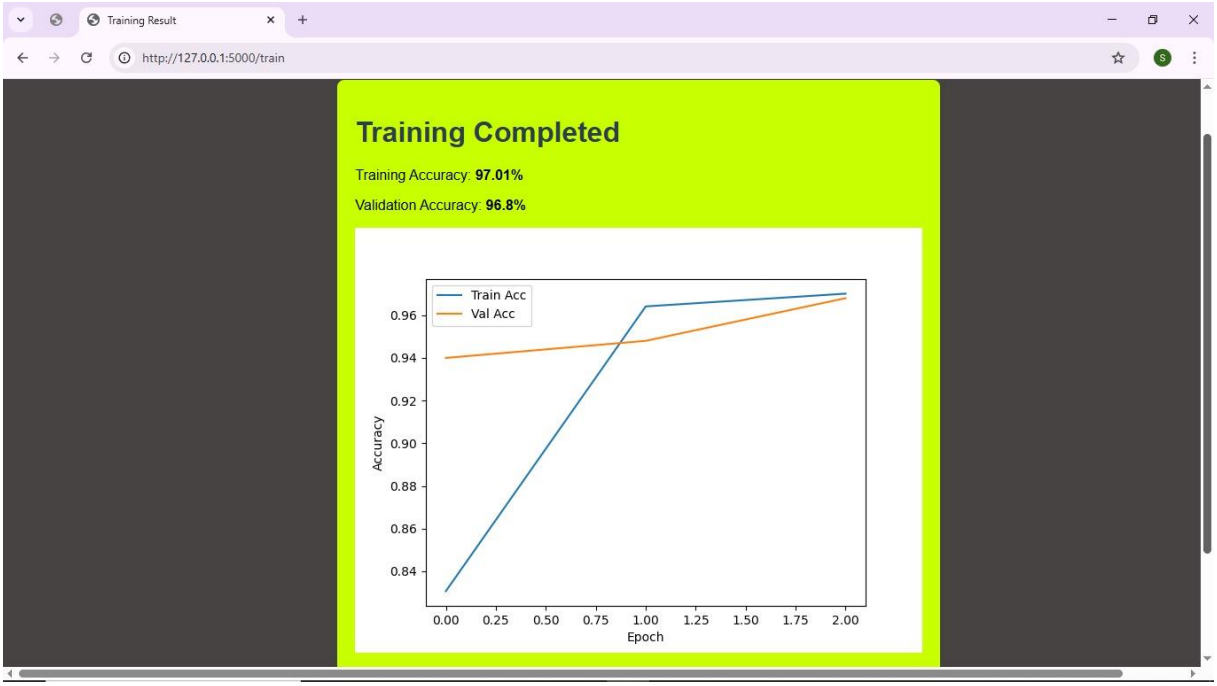
FACE MASK DETECTION



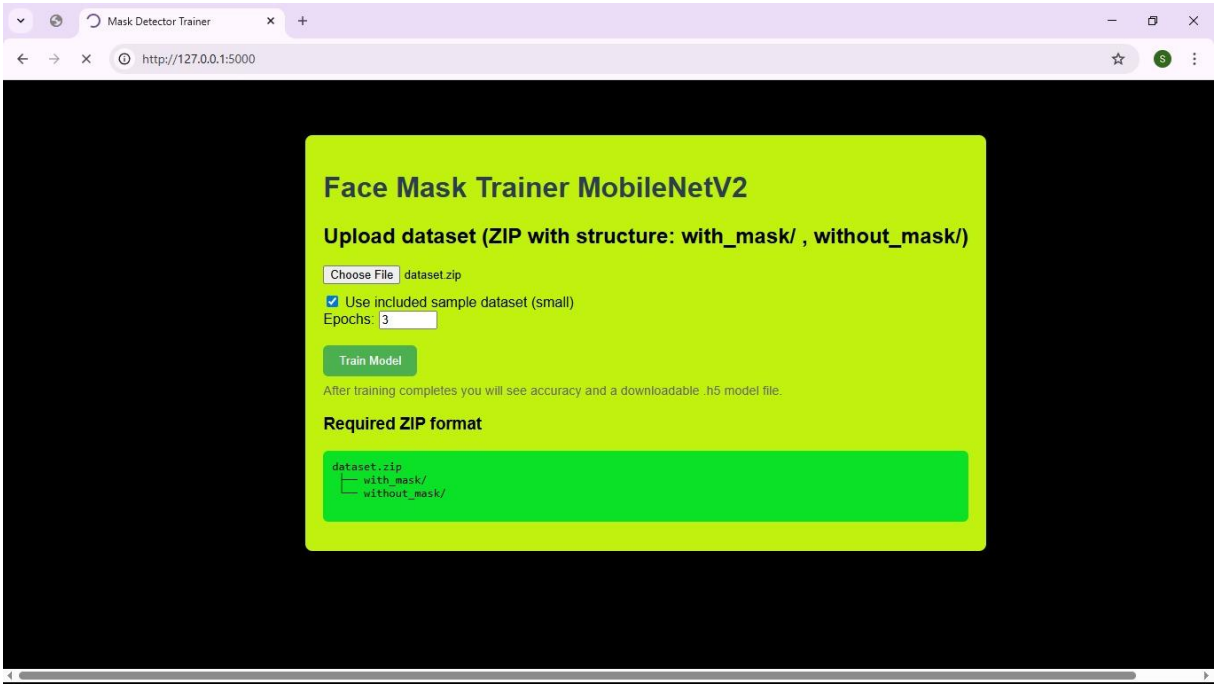
CNN HOME PAGE



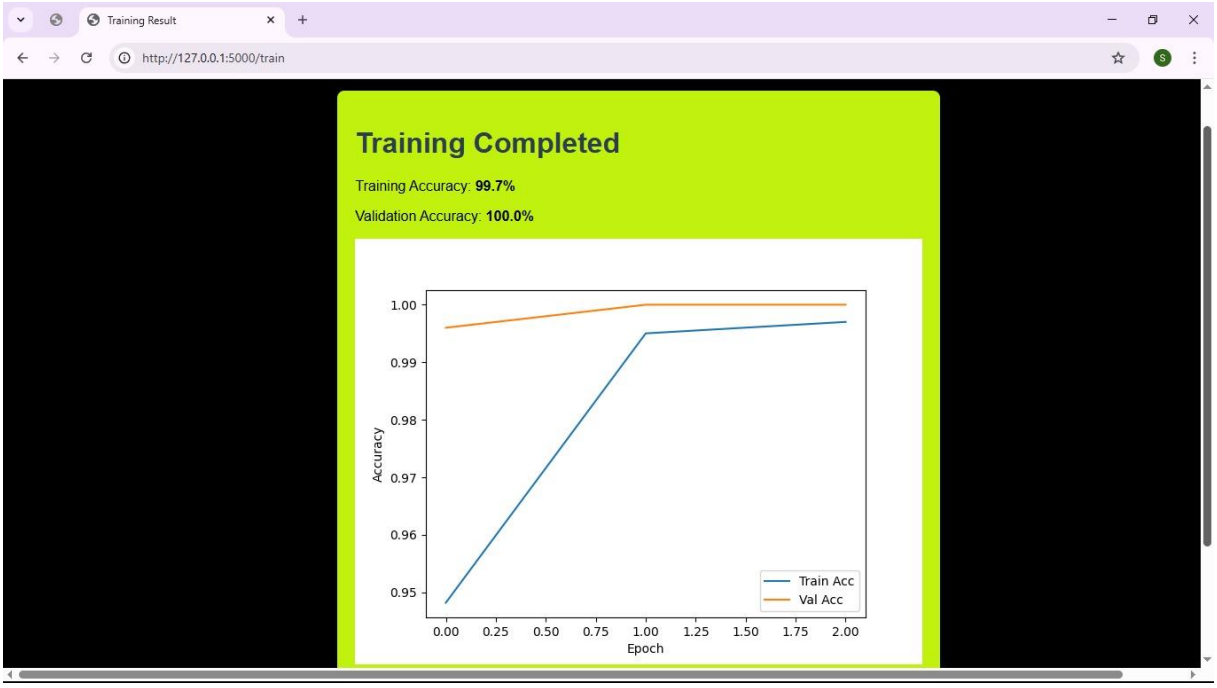
CNN ACCURACY PAGE



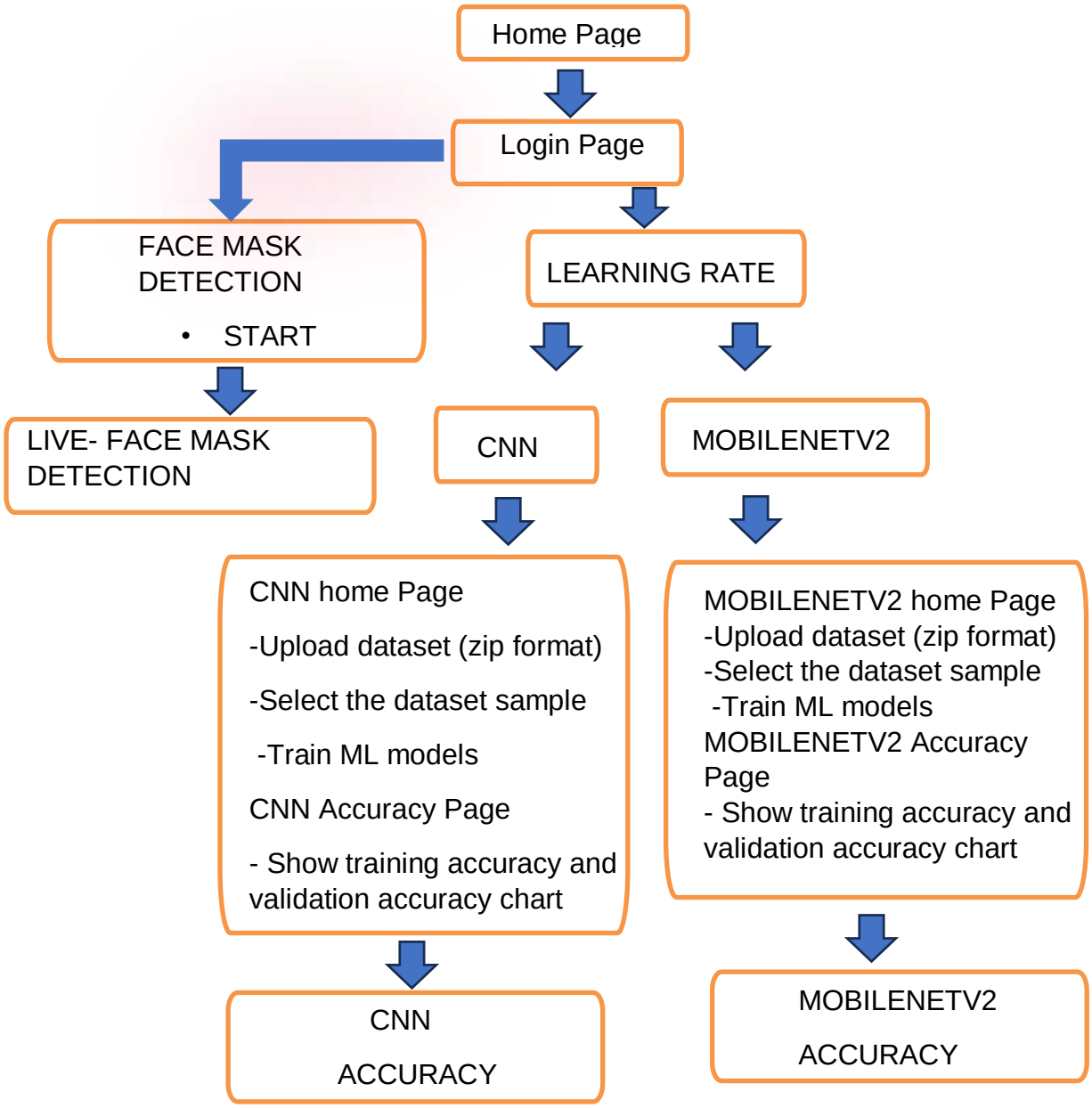
MOBILENETV2 HOME PAGE



MOBILENETV2 ACCURACY PAGE



5.3 System Design



CHAPTER-5

6.1 LIVE- FACE MASK DETECTION CODING

```
from flask import Flask, Response, render_template
import cv2
import numpy as np
from tensorflow.keras.models import load_model
from tensorflow.keras.preprocessing.image import img_to_array
from tensorflow.keras.applications.mobilenet_v2 import preprocess_input
import matplotlib.pyplot as plt
import io, base64

app = Flask(__name__)

# Load face detector
faceNet = cv2.dnn.readNet("face_detector/deploy.prototxt",
                          "face_detector/res10_300x300_ssd_iter_140000.caffemodel")

# Load mask detector model
maskNet = load_model("mask_detector.keras")

accuracy_records = []

# If training plot exists
training_plot_data = None

def detect_and_predict_mask(frame):
    (h, w) = frame.shape[:2]
```

```
    blob = cv2.dnn.blobFromImage(frame, 1.0, (224,224),
                                 (104.0, 177.0, 123.0))
    faceNet.setInput(blob)
    detections = faceNet.forward()
    faces, locs = [], []
    for i in range(0, detections.shape[2]):
        conf = detections[0,0,i,2]
        if conf > 0.5:
            box = detections[0,0,i,3:7]*np.array([w,h,w,h])
            (startX,startY,endX,endY) = box.astype("int")
            (startX,startY) = max(0,startX), max(0,startY)
            (endX,endY) = min(w-1,endX), min(h-1,endY)
            face = frame[startY:endY, startX:endX]
            if face.size == 0: continue
            face = cv2.cvtColor(face, cv2.COLOR_BGR2RGB)
            face = cv2.resize(face,(224,224))
            face = img_to_array(face)
            face = preprocess_input(face)
            faces.append(face)
            locs.append((startX,startY,endX,endY))
    preds=[]
    if len(faces) > 0:
        faces=np.array(faces,dtype="float32")
        preds=maskNet.predict(faces,batch_size=32)
    return (locs,preds)

def generate_frames():
    global accuracy_records
    camera = cv2.VideoCapture(0)
    while True:
```

```
        success, frame = camera.read()
        if not success: break
        frame = cv2.resize(frame,(640,480))
        (locs,preds) = detect_and_predict_mask(frame)
        frame_records=[]
        for (box,pred) in zip(locs,preds):
            (startX,startY,endX,endY)=box
            (mask,withoutMask)=pred
            label="Mask" if mask>withoutMask else "No Mask"
            color=(0,255,0) if label=="Mask" else (0,0,255)
            text="{}: {:.2f}%".format(label,max(mask,withoutMask)*100)
            cv2.putText(frame,text,(startX,startY-10),
                        cv2.FONT_HERSHEY_SIMPLEX,0.45,color,2)
            cv2.rectangle(frame,(startX,startY),(endX,endY),color,2)
            frame_records.append({"Model":label,"Accuracy":float(max(mask,withoutMask))})
        accuracy_records = frame_records
        ret,buffer=cv2.imencode('.jpg',frame)
        frame_bytes = buffer.tobytes()
        yield (b'--frame\r\n'
              b'Content-Type: image/jpeg\r\n\r\n'+frame_bytes+b'\r\n')

@app.route('/')
def index():
    global accuracy_records
    return render_template('index.html', accuracies=accuracy_records)

@app.route('/video_feed')
def video_feed():
    return Response(generate_frames(),
                    mimetype='multipart/x-mixed-replace; boundary=frame')
```

```
# Route for CNN training plot
@app.route('/train_plot')
def train_plot():
    global training_plot_data
    if training_plot_data is None:
        return "<h2>No training plot available. Train your model first.</h2>"
    return render_template('train_plot.html', plot_data=training_plot_data)

# Function to generate training plot (call after training)
def generate_training_plot(H):
    global training_plot_data
    plt.figure()
    plt.plot(H.history["accuracy"], label="train_acc")
    plt.plot(H.history["val_accuracy"], label="val_acc")
    plt.plot(H.history["loss"], label="train_loss")
    plt.plot(H.history["val_loss"], label="val_loss")
    plt.title("Training Accuracy & Loss")
    plt.xlabel("Epoch")
    plt.ylabel("Value")
    plt.legend()
    plt.tight_layout()
    buf = io.BytesIO()
    plt.savefig(buf, format='png')
    buf.seek(0)
    training_plot_data = base64.b64encode(buf.getvalue()).decode('utf-8')
    plt.close()

if __name__=="__main__":
    app.run(debug=True)
```

6.2 CNN CODING

```
from flask import Flask, render_template, request, redirect, url_for, send_file
import os, zipfile, shutil, io, time
import numpy as np
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt

import tensorflow as tf
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense, Dropout

app = Flask(__name__)
app.config['UPLOAD_FOLDER'] = 'uploads'
app.config['MODEL_FOLDER'] = 'models'
os.makedirs(app.config['UPLOAD_FOLDER'], exist_ok=True)
os.makedirs(app.config['MODEL_FOLDER'], exist_ok=True)

@app.route('/')
def index():
    return render_template('index.html')

@app.route('/train', methods=['POST'])
def train():
    # Accept either a zip upload or use the included sample dataset
    use_sample = request.form.get('use_sample') == 'on'
    if use_sample:
```

```
    # If using sample, copy included sample_data to uploads/data
    sample_src = os.path.join(os.getcwd(), 'sample_data')
    extract_path = os.path.join(app.config['UPLOAD_FOLDER'], 'data')
    if os.path.exists(extract_path):
        shutil.rmtree(extract_path)
    shutil.copytree(sample_src, extract_path)
else:
    if 'dataset' not in request.files:
        return 'No file uploaded', 400
    file = request.files['dataset']
    if file.filename == '':
        return 'No file selected', 400
    zip_path = os.path.join(app.config['UPLOAD_FOLDER'], file.filename)
    file.save(zip_path)
    extract_path = os.path.join(app.config['UPLOAD_FOLDER'], 'data')
    if os.path.exists(extract_path):
        shutil.rmtree(extract_path)
    os.makedirs(extract_path, exist_ok=True)
    with zipfile.ZipFile(zip_path, 'r') as zip_ref:
        zip_ref.extractall(extract_path)

# Expect structure: data/with_mask/*  data/without_mask/*
data_dir = os.path.join(app.config['UPLOAD_FOLDER'], 'data')
if not os.path.exists(data_dir):
    return "No dataset found after extraction.", 400

img_height, img_width = 128, 128
batch_size = 16

datagen = ImageDataGenerator(validation_split=0.2, rescale=1./255)
```

```
train_data = datagen.flow_from_directory(
    data_dir,
    target_size=(img_height, img_width),
    batch_size=batch_size,
    class_mode='binary',
    subset='training',
    shuffle=True
)

val_data = datagen.flow_from_directory(
    data_dir,
    target_size=(img_height, img_width),
    batch_size=batch_size,
    class_mode='binary',
    subset='validation',
    shuffle=False
)

# Simple CNN model
model = Sequential([
    Conv2D(32, (3,3), activation='relu', input_shape=(img_height,img_width,3)),
    MaxPooling2D(2,2),
    Conv2D(64,(3,3),activation='relu'),
    MaxPooling2D(2,2),
    Conv2D(128,(3,3),activation='relu'),
    MaxPooling2D(2,2),
    Flatten(),
    Dense(128,activation='relu'),
    Dropout(0.5),
```

```
        Dense(1,activation='sigmoid')
    ])

    model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])

    epochs = int(request.form.get('epochs', 3))
    history = model.fit(train_data, validation_data=val_data, epochs=epochs)

    acc = history.history['accuracy'][-1]
    val_acc = history.history['val_accuracy'][-1]

    # Save model as .h5
    timestamp = int(time.time())
    model_path = os.path.join(app.config['MODEL_FOLDER'],
f"mask_detector_{timestamp}.h5")
    model.save(model_path)

    # Plot accuracy graph
    plt.figure()
    plt.plot(history.history['accuracy'], label='Train Acc')
    plt.plot(history.history['val_accuracy'], label='Val Acc')
    plt.xlabel('Epoch')
    plt.ylabel('Accuracy')
    plt.legend()
    graph_path = os.path.join('static','acc.png')
    plt.savefig(graph_path)
    plt.close()

    return render_template('result.html', acc=round(acc*100,2), val_acc=round(val_acc*100,2),
graph='acc.png', model_file=os.path.basename(model_path))
```

```
@app.route('/download/<model_file>')
def download_model(model_file):
    path = os.path.join(app.config['MODEL_FOLDER'], model_file)
    if os.path.exists(path):
        return send_file(path, as_attachment=True)
    return "Model not found", 404

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000, debug=True)
```

6.3 MOBILENETV2 CODING

```
from flask import Flask, render_template, request, send_file
import os, zipfile, shutil, time
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt

import tensorflow as tf
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.layers import GlobalAveragePooling2D, Dense, Dropout
from tensorflow.keras.models import Model

app = Flask(__name__)
app.config['UPLOAD_FOLDER'] = 'uploads'
app.config['MODEL_FOLDER'] = 'models'
os.makedirs(app.config['UPLOAD_FOLDER'], exist_ok=True)
os.makedirs(app.config['MODEL_FOLDER'], exist_ok=True)
```

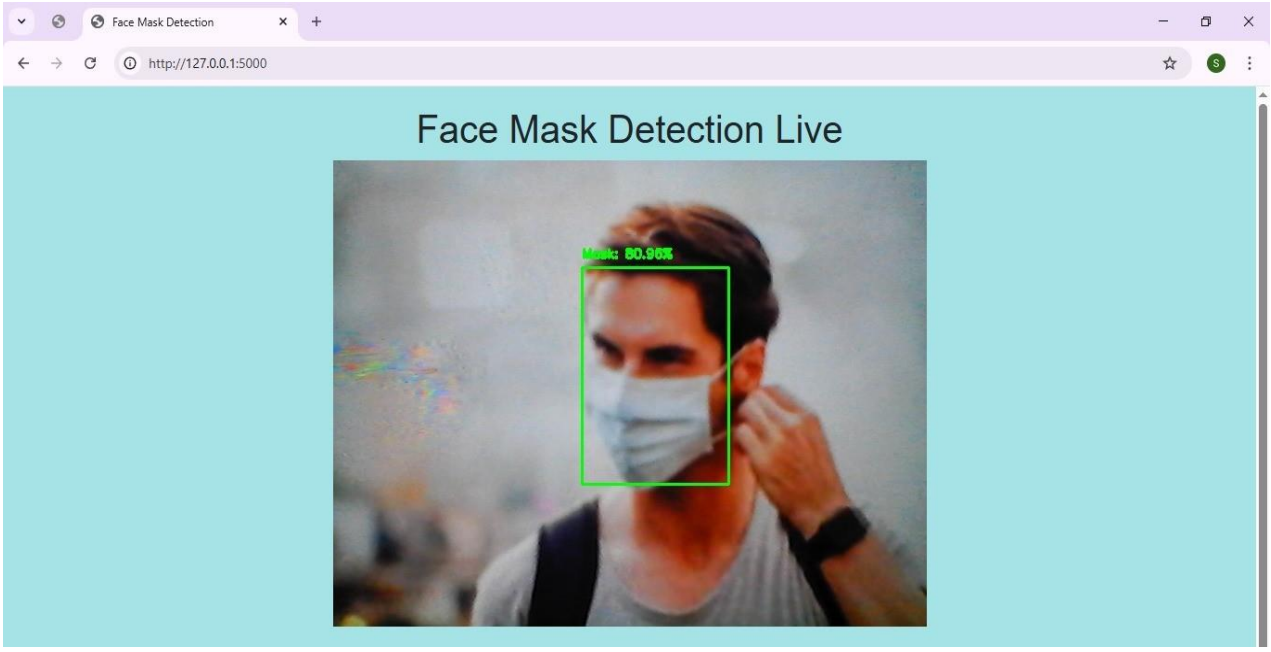
```
@app.route('/')
def index():
    return render_template('index.html')

@app.route('/train', methods=['POST'])
def train():
    # Use sample dataset or uploaded dataset
    use_sample = request.form.get('use_sample') == 'on'
    if use_sample:
        sample_src = os.path.join(os.getcwd(), 'sample_data')
        extract_path = os.path.join(app.config['UPLOAD_FOLDER'], 'data')
        if os.path.exists(extract_path):
            shutil.rmtree(extract_path)
        shutil.copytree(sample_src, extract_path)
    else:
        if 'dataset' not in request.files:
            return 'No file uploaded', 400
        file = request.files['dataset']
        if file.filename == '':
            return 'No file selected', 400
        zip_path = os.path.join(app.config['UPLOAD_FOLDER'], file.filename)
        file.save(zip_path)
        extract_path = os.path.join(app.config['UPLOAD_FOLDER'], 'data')
        if os.path.exists(extract_path):
            shutil.rmtree(extract_path)
        os.makedirs(extract_path, exist_ok=True)
        with zipfile.ZipFile(zip_path, 'r') as zip_ref:
            zip_ref.extractall(extract_path)

    # Dataset directory
```

```
data_dir = os.path.join(app.config['UPLOAD_FOLDER'], 'data')
if not os.path.exists(data_dir):
    return "No dataset found after extraction.", 400

img_height, img_width = 224, 224  # MobileNetV2 requirement
batch_size = 16

datagen = ImageDataGenerator(
    validation_split=0.2,
    preprocessing_function=tf.keras.applications.mobilenet_v2.preprocess_input
)

train_data = datagen.flow_from_directory(
    data_dir,
    target_size=(img_height, img_width),
    batch_size=batch_size,
    class_mode='binary',
    subset='training',
    shuffle=True
)

val_data = datagen.flow_from_directory(
    data_dir,
    target_size=(img_height, img_width),
    batch_size=batch_size,
    class_mode='binary',
    subset='validation',
    shuffle=False
)
```

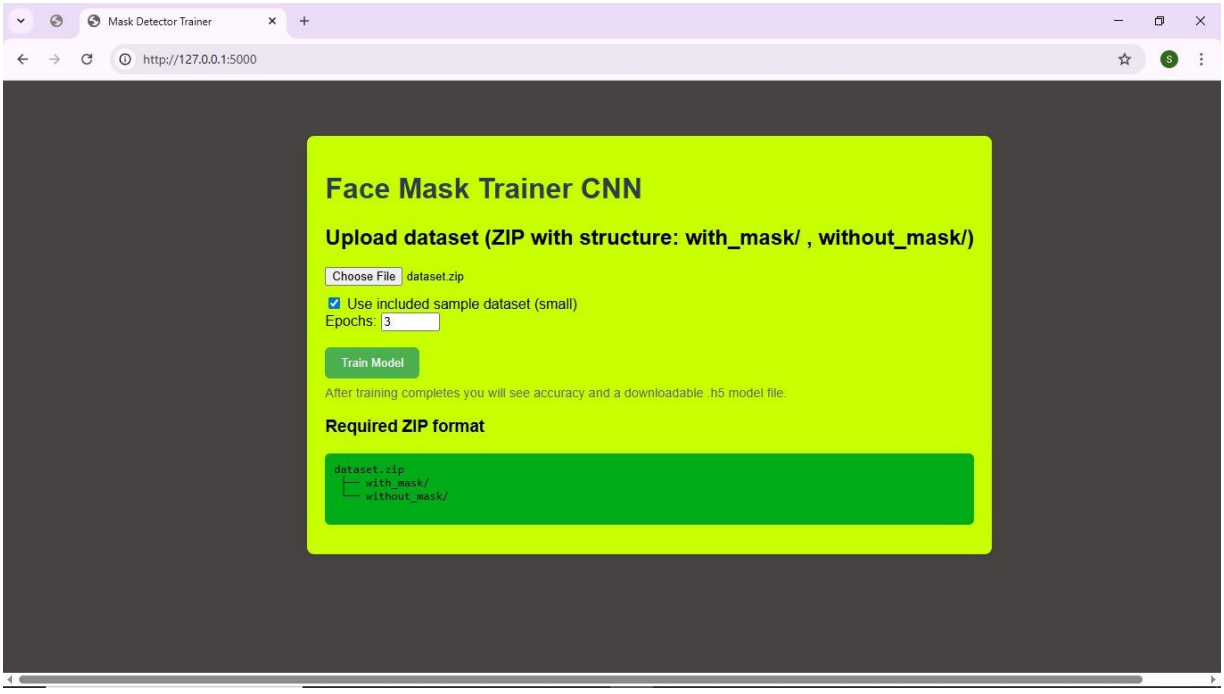
```
# --- MobileNetV2 model ---
base_model = MobileNetV2(weights='imagenet', include_top=False,
                         input_shape=(img_height, img_width, 3))
base_model.trainable = False  # freeze base layers

x = base_model.output
x = GlobalAveragePooling2D()(x)
x = Dense(128, activation='relu')(x)
x = Dropout(0.5)(x)
preds = Dense(1, activation='sigmoid')(x)

model = Model(inputs=base_model.input, outputs=preds)
model.compile(optimizer='adam',
              loss='binary_crossentropy',
              metrics=['accuracy'])

# Train model
epochs = int(request.form.get('epochs', 3))
history = model.fit(train_data, validation_data=val_data, epochs=epochs)

acc = history.history['accuracy'][-1]
val_acc = history.history['val_accuracy'][-1]

# Save model
timestamp = int(time.time())
model_path = os.path.join(app.config['MODEL_FOLDER'],
f"mask_detector_mobilenetv2_{timestamp}.h5")
model.save(model_path)

# Accuracy graph
plt.figure()
```

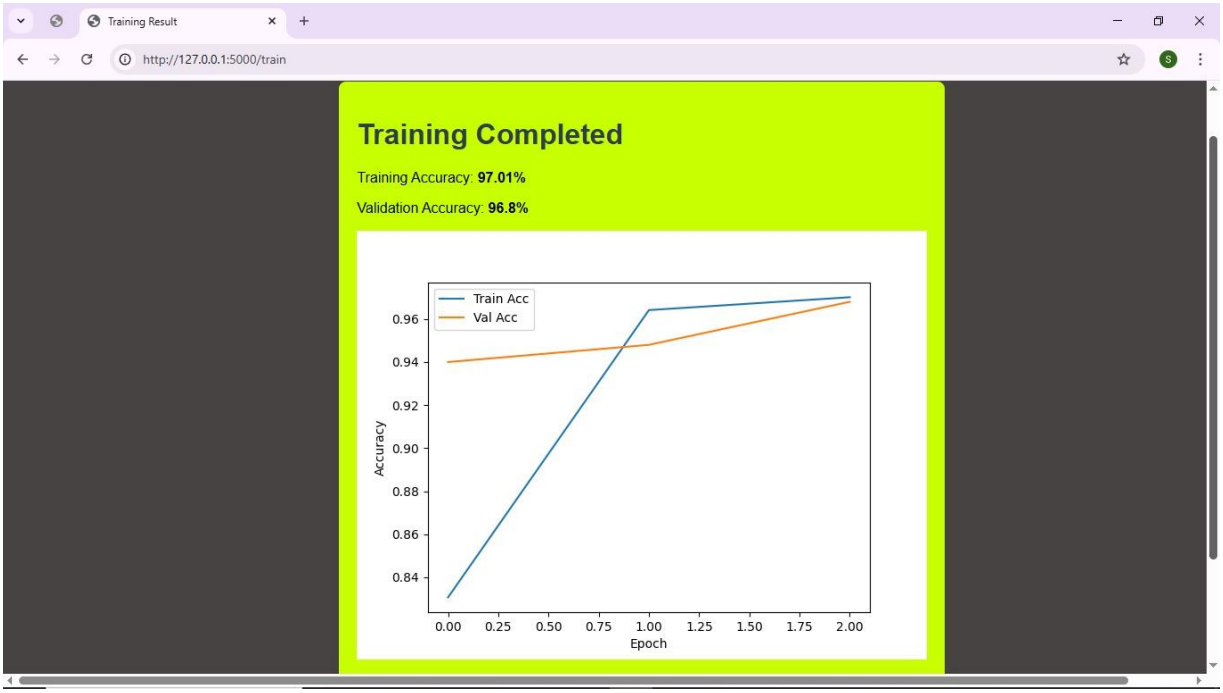
```
    plt.plot(history.history['accuracy'], label='Train Acc')
    plt.plot(history.history['val_accuracy'], label='Val Acc')
    plt.xlabel('Epoch')
    plt.ylabel('Accuracy')
    plt.legend()
    graph_path = os.path.join('static', 'acc.png')
    plt.savefig(graph_path)
    plt.close()

    return render_template(
        'result.html',
        acc=round(acc*100, 2),
        val_acc=round(val_acc*100, 2),
        graph='acc.png',
        model_file=os.path.basename(model_path)
    )

@app.route('/download/<model_file>')
def download_model(model_file):
    path = os.path.join(app.config['MODEL_FOLDER'], model_file)
    if os.path.exists(path):
        return send_file(path, as_attachment=True)
    return "Model not found", 404

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000, debug=False)
```

6.1.1 RESULT

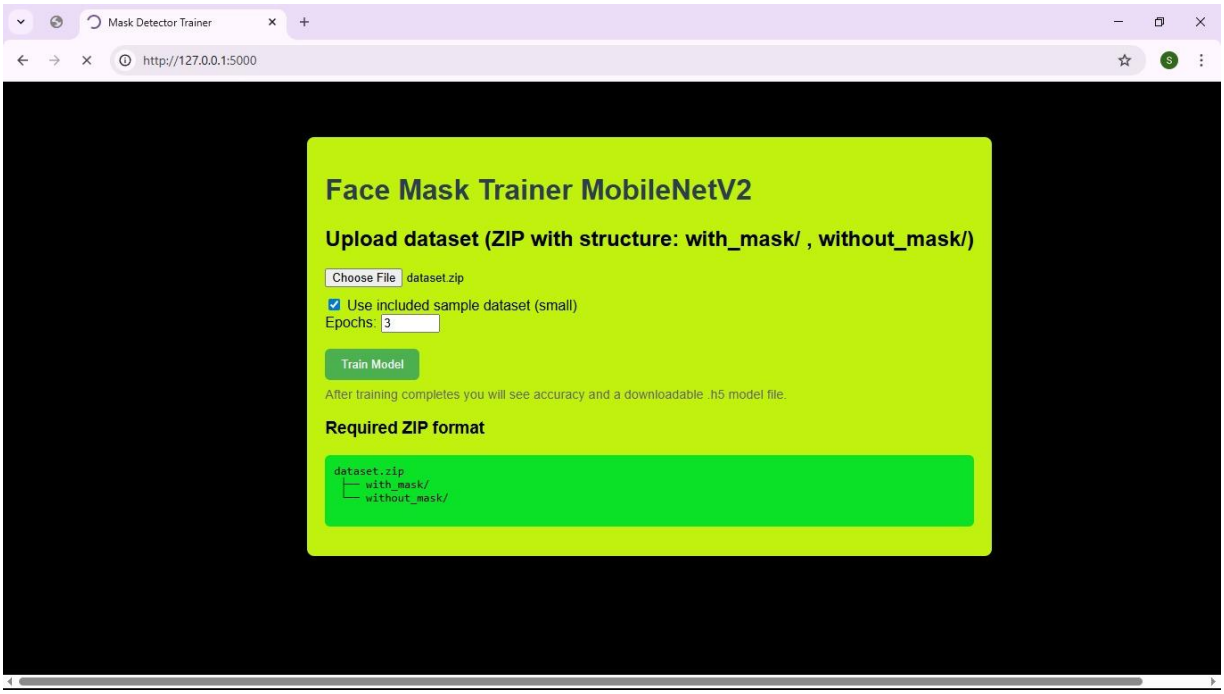


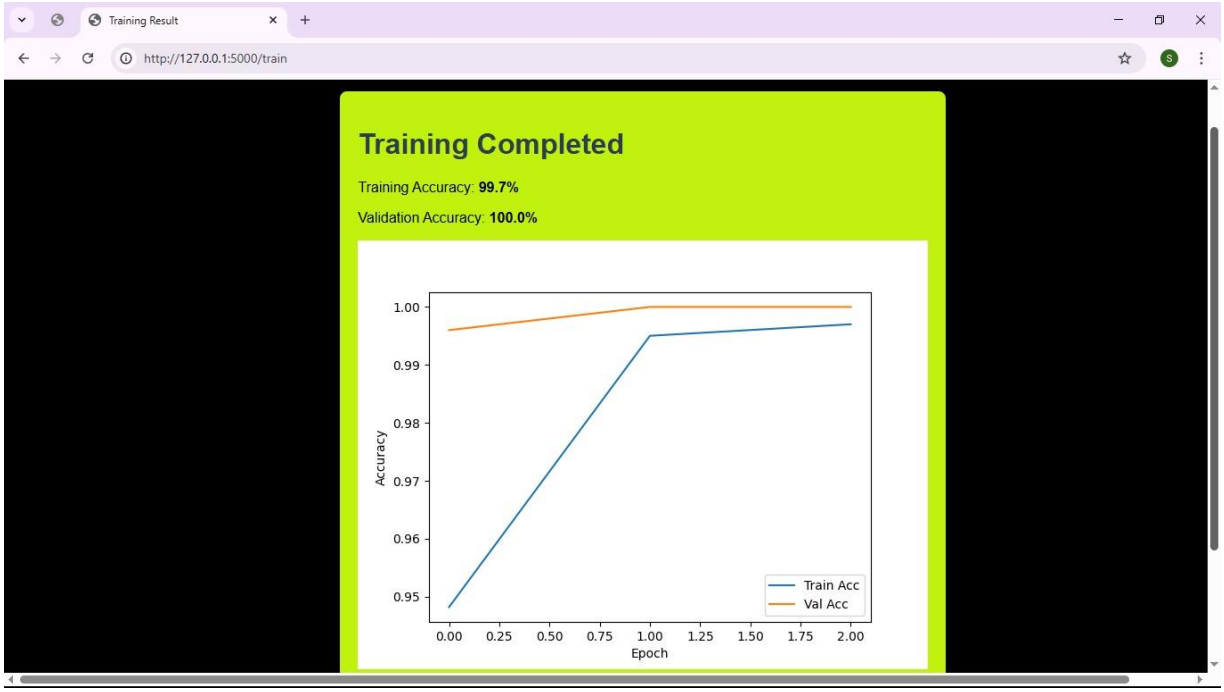
6.2.1 RESULT




Training Result
http://127.0.0.1:5000/train
Training Completed
Training Accuracy: 97.01%
Validation Accuracy: 96.8%
Train Acc
Val Acc
Accuracy
Epoch
0.96
0.94
0.92
0.90
0.88
0.86
0.84
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00

6.3.1 RESULT


Mask Detector Trainer
http://127.0.0.1:5000
Face Mask Trainer MobileNetV2
Upload dataset (ZIP with structure: with_mask/ , without_mask/)
Choose File
dataset.zip
Use included sample dataset (small)
Epochs:
3
Train Model
After training completes you will see accuracy and a downloadable .h5 model file.
Required ZIP format
dataset.zip
├── with_mask/
└── without_mask/


Training Result
http://127.0.0.1:5000/train
Training Completed
Training Accuracy: 99.7%
Validation Accuracy: 100.0%
Accuracy
Epoch
Train Acc
Val Acc

CHAPTER-6

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The face mask detection system developed in this study successfully demonstrates the application of deep learning techniques to solve a real-world public health challenge. By leveraging CNN and MobileNetV2 architectures, the system achieves high accuracy in identifying whether individuals are wearing face masks, even under varied lighting conditions and facial orientations. This confirms the effectiveness of AI-based solutions in automating safety protocols.

The modular design of the system ensures flexibility and scalability. Each component—from image input to classification and alert generation—functions independently, allowing for easy integration into existing surveillance or access control infrastructures. This structure also supports future upgrades without requiring a complete system overhaul.

The use of a diverse and augmented dataset contributes significantly to the model's generalization capability. By including images with different demographics, mask types, and environmental conditions, the system avoids bias and performs reliably across a wide range of scenarios. This enhances its suitability for deployment in public spaces such as hospitals, schools, and transportation hubs.

Evaluation metrics such as accuracy, precision, recall, and F1-score validate the robustness of the model. The confusion matrix analysis further highlights the system's strengths and areas

for improvement, guiding future refinements. The results confirm that the model is not only technically sound but also practically viable for real-time monitoring.

Overall, this research contributes to the growing field of intelligent health monitoring systems. It bridges the gap between public health policy and technological enforcement, offering a scalable, contactless, and efficient solution for mask compliance. The study lays a strong foundation for future innovations in computer vision applications for healthcare.

5.2 Future Scope

While the current system focuses on binary classification—detecting whether a mask is worn or not—future work can expand its capabilities to include detection of improper mask usage. This would involve training the model to recognize masks worn below the nose, loosely fitted masks, or other non-compliant behaviors. Such enhancements would make the system more aligned with evolving public health standards.

Another promising direction is the integration of the mask detection system with other intelligent monitoring tools. For example, combining it with thermal imaging for fever detection or crowd density analysis could create a comprehensive health surveillance platform. This would be particularly useful in high-risk environments like airports and large public gatherings.

The system can also benefit from advancements in edge computing and federated learning. Deploying the model on edge devices would reduce latency and improve privacy, while federated learning would allow the system to learn from distributed data sources without compromising user confidentiality. These technologies can make the system more adaptive and secure.

Expanding the dataset to include more real-world scenarios, such as outdoor environments, different cultural contexts, and various mask designs, would further improve the model's robustness. Collaborations with public health organizations could facilitate access to such data and ensure that the system remains relevant and effective.

Future research can explore the ethical and social implications of automated mask detection. Ensuring transparency, data privacy, and responsible use will be critical as such systems become more widespread. Engaging with stakeholders—including policymakers, healthcare professionals, and the public—will help guide the responsible evolution of this technology.

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